

Technology Stack

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	Provides a responsive and user-friendly interface for entering weather parameters and displaying flood prediction results.
2	Backend / Server-Side	Python, Flask	Flask is lightweight and integrates easily with machine learning models for real-time flood prediction.
3	Database / Data Storage	Excel Dataset (.xlsx), Joblib (.pkl) Model	Historical flood dataset is stored in Excel format, while the trained machine learning model is saved as a Joblib (.pkl) file for fast prediction.
4	Cloud / Hosting / Deployment	PythonAnywhere	Used to deploy the Flask web application online with free hosting for easy access.
5	Version Control & CI/CD	Git, GitHub	Enables version control, project backup, and team collaboration throughout development.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, OpenPyXL, Joblib	Scikit-learn trains the prediction model, Pandas and NumPy handle data processing, OpenPyXL reads Excel datasets, and Joblib loads the trained model efficiently.